

A Sharp Logarithmic Almost-Sure Threshold for a Critical Two-Step Pearson Walk

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Abstract

For each dimension $d \geq 2$, consider two independent uniform directions on the unit sphere in $\mathbb{R}^d$ and independent step lengths 1 and $\sqrt{d}(\log d)^{-\alpha} X_d$, where $X_d \sim \text{Ber}(d^{-1/2})$. The triples of random inputs are assumed independent across dimensions. We determine exactly when the squared norm of the resulting two-step Pearson walk differs from the squared norm of its length vector by a quantity tending to zero almost surely. For each fixed $\alpha \in \mathbb{R}$, convergence holds if and only if $\alpha > 1/2$. At the critical value $\alpha = 1/2$, the exceedance series at level ε converges exactly for $\varepsilon > 2$, so the endpoint itself fails almost surely. The proof is based on a two-sided moderate-tail estimate for one spherical coordinate and the two Borel–Cantelli lemmas.

1 Statement and context

On one probability space, for every integer $d \geq 2$, let

$$X_d \sim \text{Ber}(d^{-1/2}) \quad \text{and} \quad U_1^{(d)}, U_2^{(d)} \sim \text{Unif}(S^{d-1}).$$

Within each dimension these three variables are independent, and the triples are mutually independent across d . For a fixed $\alpha \in \mathbb{R}$, set

$$L_1^{(d)} = 1, \quad L_2^{(d)} = \sqrt{d}(\log d)^{-\alpha} X_d, \quad S_{d,2} = L_1^{(d)} U_1^{(d)} + L_2^{(d)} U_2^{(d)},$$

and define

$$Y_d = \|S_{d,2}\|_2^2 - ((L_1^{(d)})^2 + (L_2^{(d)})^2).$$

The probability-one statement below is separate for each fixed α ; no common null set over all real parameters is asserted.

Theorem 1. *For each fixed $\alpha \in \mathbb{R}$,*

$$Y_d \longrightarrow 0 \quad \text{almost surely} \quad \Longleftrightarrow \quad \alpha > \frac{1}{2}.$$

More precisely, for every $\varepsilon > 0$, let $E_d(\varepsilon) = \{|Y_d| > \varepsilon\}$. Then

$$\sum_{d=2}^{\infty} \mathbb{P}(E_d(\varepsilon)) \begin{cases} \text{diverges for every } \varepsilon > 0, & \alpha < 1/2, \\ \text{converges exactly when } \varepsilon > 2, & \alpha = 1/2, \\ \text{converges for every } \varepsilon > 0, & \alpha > 1/2. \end{cases}$$

High-dimensional isotropic Pearson walks and their concentration properties are studied systematically by Bignamini, Casini, and Martinelli [1]. Their Example 6.5 treats the neighboring polynomial family in which the random second length is $d^\beta X_d$, including failure at the critical power $\beta = 1/2$. The calculation below supplies the logarithmically damped classification at that critical power and resolves its endpoint. We make no claim of priority.

2 A spherical moderate-tail estimate

We write $A_d \asymp B_d$ when both ratios are bounded above by positive constants independent of all sufficiently large d ; the constants may depend on fixed parameters.

Lemma 1. *Let V_d have the law of one coordinate of a uniform point on S^{d-1} and put $Z_d = \sqrt{d} V_d$. For fixed $c, \gamma > 0$,*

$$\mathbb{P}(|Z_d| > c(\log d)^\gamma) \asymp (\log d)^{-\gamma} \exp\left[-\frac{c^2}{2}(\log d)^{2\gamma}\right]. \quad (1)$$

If a nonnegative threshold t_d is eventually bounded above, then $\mathbb{P}(|Z_d| > t_d)$ is bounded below by a positive constant for all sufficiently large d .

Proof. The density of Z_d is

$$h_d(z) = a_d \left(1 - \frac{z^2}{d}\right)^{(d-3)/2} \mathbf{1}_{\{|z| < \sqrt{d}\}}, \quad a_d = \frac{\Gamma(d/2)}{\sqrt{\pi d} \Gamma((d-1)/2)}. \quad (2)$$

Stirling's formula gives $a_d \rightarrow (2\pi)^{-1/2}$, so these normalizing constants are bounded above and away from zero.

Put $t = c(\log d)^\gamma$. Then $t^2/d \rightarrow 0$ and $t^4/d \rightarrow 0$. The inequality $\log(1-u) \leq -u$, followed by the standard Gaussian tail bound, gives

$$\begin{aligned} \mathbb{P}(|Z_d| > t) &\leq 2a_d \int_t^\infty \exp\left(-\frac{d-3}{2d}z^2\right) dz \\ &\leq Ct^{-1} \exp(-t^2/2) \exp(3t^2/(2d)) \leq C't^{-1} \exp(-t^2/2). \end{aligned}$$

For the reverse inequality, integrate only over $[t, t+t^{-1}]$. Eventually this interval lies inside the support in (2), $t \geq 1$, and $u = z^2/d \leq 1/2$. Since $\log(1-u) \geq -u - u^2$ on $[0, 1/2]$ and the logarithm is nonpositive,

$$\frac{d-3}{2} \log(1 - z^2/d) \geq -\frac{z^2}{2} - \frac{z^4}{2d}.$$

Uniformly on the chosen interval, $z^2 \leq t^2 + 3$ and $z^4/d = o(1)$. Multiplication by the interval length t^{-1} therefore yields

$$\mathbb{P}(|Z_d| > t) \geq c_0 t^{-1} e^{-t^2/2},$$

which proves (1). Finally, if $t_d \leq M$ eventually, integration over any fixed interval strictly above M gives a uniform positive lower bound by the same density estimates. $\square$

3 Proof of the threshold

By rotational invariance,

$$V_d = \langle U_1^{(d)}, U_2^{(d)} \rangle$$

has the coordinate law used in Lemma 1. Direct expansion gives the exact identity

$$Y_d = 2\sqrt{d}(\log d)^{-\alpha} X_d V_d = 2(\log d)^{-\alpha} X_d Z_d. \quad (3)$$

Independence of X_d and the directions consequently implies

$$\mathbb{P}(E_d(\varepsilon)) = d^{-1/2} \mathbb{P}\left(|Z_d| > \frac{\varepsilon}{2}(\log d)^\alpha\right). \quad (4)$$

If $\alpha \leq 0$, the threshold in (4) is eventually bounded, and Lemma 1, together with the trivial upper bound, gives $\mathbb{P}(E_d(\varepsilon)) \asymp d^{-1/2}$. Its series diverges.

Now suppose $\alpha > 0$. Applying (1) with $c = \varepsilon/2$ and $\gamma = \alpha$ yields

$$\mathbb{P}(E_d(\varepsilon)) \asymp d^{-1/2}(\log d)^{-\alpha} \exp\left[-\frac{\varepsilon^2}{8}(\log d)^{2\alpha}\right]. \quad (5)$$

For $0 < \alpha < 1/2$, the exponential factor is $d^{-o(1)}$. For instance, eventually it is at least $d^{-1/4}$, while $(\log d)^{-\alpha} \geq d^{-1/8}$. Thus the terms in (5) are bounded below by a constant multiple of $d^{-7/8}$, and the series diverges. For $\alpha > 1/2$, the exponential factor is eventually at most d^{-1} ; the resulting upper bound is summable.

At $\alpha = 1/2$, formula (5) becomes

$$\mathbb{P}(E_d(\varepsilon)) \asymp d^{-1/2-\varepsilon^2/8}(\log d)^{-1/2}. \quad (6)$$

The power-logarithm series criterion shows convergence exactly when $\varepsilon > 2$. In particular, at $\varepsilon = 2$ the terms are comparable to $1/(d\sqrt{\log d})$, whose series diverges.

It remains to translate these series statements into the almost-sure claim. When $\alpha > 1/2$, the first Borel–Cantelli lemma applied successively at the countable levels $\varepsilon = 1/m$, $m \geq 1$, implies $Y_d \rightarrow 0$ almost surely. When $\alpha < 1/2$, take $\varepsilon = 1$; at $\alpha = 1/2$, take the same level. The corresponding series diverges. The events are independent across d because each depends only on its dimension- d input triple, so the second Borel–Cantelli lemma gives $|Y_d| > 1$ infinitely often almost surely. This excludes convergence to zero and completes the proof of Theorem 1.

4 Scope and methodological note

Only the product-law assumptions stated above are used. In particular, within-dimension independence is needed in (4), and cross-dimensional independence is needed only for the divergent half of the Borel–Cantelli argument. Without the former assumption, couplings between X_d and the directions can change the answer.

As a methodological comparison, Zeng [2] studies a different combinatorial-probability monotonicity question. No result from that work is used here. The present proof is entirely analytic and contains no simulation or finite computer verification.

Computational and AI assistance. An AI language model assisted with organizing the audited argument, checking bibliographic metadata, and preparing the L^AT_EX manuscript. The proof was also examined in a separate model context; this is not independent peer review. No computer algebra, numerical experiment, or simulation is used as mathematical evidence.

References

- [1] Davide A. Bignamini, Emanuele G. Casini, and Andrea Martinelli. Asymptotic behaviour for isotropic pearson random walks, 2026. arXiv:2609.05195 [math.PR].
- [2] Zijian Zeng. The Distinct-Cycle-Length Probability is Strictly Decreasing after 30. SSRN preprint, 2026. SSRN 7380519.